Can "Most Underrated" Prompts Diagnose LLM Novelty? A Data-Grounded Anti-Consensus Evaluation

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Abstract

Prompts such as "most underrated X" are widely used for discovery, but it is unclear whether they reveal genuine model novelty or just recycled crowd-approved picks. We introduce a reproducible diagnostic that combines real API outputs, a data-grounded underratedness proxy from MOVIELENS-LATEST-SMALL, and social-consensus overlap signals from REDDIT. We evaluate two model families (GPT-4.1 and GPT-5), nine movie genres, three prompt conditions (baseline, anti-consensus, anti-consensus+reasoned), and two repeated runs per cell, yielding 108 model outputs. Our robustness metric, the UNS hit score, combines percentile underratedness with a direct title-hit overlap penalty. For GPT-4.1, anti-consensus prompting substantially reduces overlap-hit rate from 38.9% in baseline to 22.2% (anti-consensus) and 11.1% (anti-consensus+reasoned), with the best mean UNS hit score under anti-consensus+reasoned (0.669 vs. 0.562 baseline). However, paired permutation tests with Holm correction show no statistically significant gains after multiple-comparison adjustment. For GPT-5 in this setup, anti-consensus variants suffer severe structured-output failures, which dominate metric outcomes. These results suggest that "most underrated" prompts are promising as novelty probes, but conclusions depend strongly on metric construction quality and output-validation reliability. We release a modular pipeline and report limitations to support stronger future benchmark design.

1 Introduction

Large language models are increasingly used for ideation and discovery tasks where users explicitly ask for non-obvious answers. A common prompt form is "what is the most underrated X?" The practical question is whether models actually discover less-obvious candidates or simply repeat socially familiar "underrated" examples.

Why this matters. If models mostly echo crowd consensus under an "underrated" prompt, their value for discovery is limited. This matters for recommendation-adjacent use cases, creative brainstorming, and any workflow that needs high-quality novelty instead of safe repetition.

Gap. Prior work has established decoding degeneration and diversity limitations in open-ended generation [Holtzman et al., 2019, Vijayakumar et al., 2016]. Other work studies distributional or lexical novelty [Pillutla et al., 2021] and novelty-aware ranking [Sharma et al., 2024]. However, there is no standard benchmark for the specific behavior of "most underrated X" prompts: balancing anti-consensus novelty with plausibility and quality.

Our approach. We propose UNDERRATED NOVELTY DIAGNOSTIC, a reproducible diagnostic that integrates three signals: (1) overlap with mined "underrated" discourse on REDDIT as an obviousness proxy, (2) a data-grounded underratedness percentile from MOVIELENS-LATEST-SMALL, and (3) output validity/matching checks. We run real API calls for GPT-4.1 and GPT-5 across 9 genres, 3 prompt conditions, and 2 runs per condition.

Quantitative preview. In robustness analysis, GPT-4.1 shows a clear directional overlap drop under anti-consensus prompting: 38.9% overlap-hit rate in baseline, 22.2% in anti-consensus, and 11.1% in anti-consensus+reasoned. The best mean robustness composite score is also from anti-consensus+reasoned (0.669). Yet none of the pairwise UNS improvements remains significant after Holm correction.

Our contributions are:

- We formulate a practical "underrated novelty" diagnostic that explicitly combines anti-consensus and data-grounded underratedness criteria.
- We conduct controlled API experiments across models, genres, and prompt interventions with paired statistical testing and multiple-comparison correction.
- We provide a robustness analysis showing directional gains in overlap reduction but limited corrected statistical significance in this pilot scale.
- We identify concrete benchmark risks, including noisy consensus mining and structured-output fragility, and provide a reproducible artifact set for follow-up work.

The rest of the paper is organized as follows. Section 2 positions our approach in prior novelty and diversity research. Section 3 describes data construction and evaluation. Section 4 reports quantitative findings, followed by implications and limitations in Section 5.

2 Related Work

Decoding degeneration and diversity. Neural text generation studies show that high-probability decoding can collapse into bland or repetitive outputs [Holtzman et al., 2019]. Diversity-aware decoding such as diverse beam search mitigates this by explicitly penalizing candidate similarity [Vijayakumar et al., 2016]. These findings motivate our hypothesis that baseline "underrated" prompts may default to socially safe answers.

Novelty and creativity interventions. Recent approaches improve novelty at inference time through candidate diversification and post-hoc judging [Franceschelli and Musolesi, 2024], or through multi-view prompting without model retraining [Lagzian et al., 2025]. We test a simpler intervention family: anti-consensus instructions with and without reasoned justification constraints.

Novelty evaluation metrics. Distribution-level metrics such as MAUVE [Pillutla et al., 2021] and lexical diversity indicators are useful but do not directly measure whether an answer is an "obvious underrated pick." Recommendation work formalizes novelty@k and relevance trade-offs [Sharma et al., 2024], while serendipity frameworks emphasize unexpected yet useful results [Xi et al., 2025, Yong et al., 2025]. Our metric design follows this logic by separating anti-consensus behavior from quality proxies.

Benchmark construction for novelty detection. Emerging work on scientific novelty detection highlights the challenge of building robust novelty ground truth [Liu et al., 2025]. Our setting has the same issue: social consensus data are noisy and incomplete. Unlike prior general novelty metrics, we target a concrete user-facing prompt pattern and explicitly report robustness gaps in consensus-ground-truth construction.

3 Methodology

Research question and hypothesis. We ask whether "most underrated X" prompts can diagnose novelty limitations in LLMs. Our core hypothesis is that baseline prompting yields socially obvious outputs (high overlap with popular REDDIT discourse), while anti-consensus prompting reduces this overlap and can improve composite underrated novelty.

3.1 Data Sources and Construction

We use MOVIELENS-LATEST-SMALL (latest small) with 9,742 movies and 100,836 ratings. We collect 3,600 REDDIT posts from `r/movies`, `r/MovieSuggestions`, `r/Letterboxd`, and `r/TrueFilm` using underrated+genre queries.

For each movie, we compute:

$$\text{underrated_raw} = z(\text{avg_rating}) - z(\log(1 + \text{rating_count})). \tag{1}$$

We convert this value into a percentile score (`underrated_pct`) globally and within genre.

Data quality checks report 0% missing values in core aggregates, no duplicate `movieId`, and 3,600 collected posts.

3.2 Experimental Design

This is an evaluation-only study (no model training). We run a full factorial grid:

- Models: GPT-4.1 and GPT-5 (low reasoning effort).
- Genres: 9 MovieLens genres.
- Prompt conditions: BASELINE, ANTI-CONSENSUS, ANTI-CONSENSUS+REASONED.
- Repeated runs: 2 per cell.

Total outputs: $2 \times 9 \times 3 \times 2 = 108$.

We parse strict JSON outputs (title, year, rationale), then fuzzy-match generated titles to MovieLens entries with year-aware constraints (threshold 75).

3.3 Metrics

We report four metrics:

- `underrated_pct`: percentile underratedness from MOVIELENS-LATEST-SMALL.
- `consensus_overlap`: overlap with mined subreddit consensus lists (primary, noisier).
- `overlap_hit`: binary overlap using direct query hits for "`<title> underrated movie`" with threshold 5 posts.
- Robustness composite score:

$$\text{UNS_hit} = 0.6 \cdot \texttt{underrated_pct} + 0.4 \cdot (1 - \texttt{overlap_hit}). \tag{2}$$

3.4 Statistical Analysis and Reproducibility

We use paired permutation tests and report paired Cohen's d for condition comparisons within each model. We correct for multiple comparisons with Holm adjustment at $\alpha = 0.05$.

Implementation details: Python 3.12.8; openai 2.30.0; pandas 3.0.2; numpy 2.4.4; scipy 1.17.1; statsmodels 0.14.6. Analysis seed is 42. Runtime for the main run is approximately 500 seconds. Hardware includes two RTX 3090 GPUs, though inference uses API calls rather than local model training.

3.5 Baselines and Prompt Interventions

We compare three prompt families for each genre:

- BASELINE: "What is the most underrated [genre] movie? Return one title and year."
- ANTI-CONSENSUS: instruction to avoid common crowd-cited picks.
- ANTI-CONSENSUS+REASONED: anti-consensus instruction plus short rationale requirement tied to under-recognition.

These prompt variants function as our baseline/intervention design for testing whether observed novelty behavior is prompt-sensitive.

Table 1: Robustness summary by model and prompt condition. Higher `UNS_hit Mean` is better; lower overlap and parse error are better. Best `UNS_hit Mean` per model is in **bold**.

Model	Condition	N	`UNS_hit Mean`	Overlap Hit Rate	Reddit Hits Mean	Match Rate	Parse Error Rate
GPT-4.1	Baseline	18	0.562	0.389	24.667	**0.889**	**0.000**
GPT-4.1	Anti-Consensus	18	0.542	0.222	6.722	0.778	**0.000**
GPT-4.1	Anti-Consensus+Reasoned	18	**0.669**	0.111	8.167	0.722	**0.000**
GPT-5	Baseline	18	**0.468**	0.333	11.278	**0.667**	0.167
GPT-5	Anti-Consensus	18	0.421	**0.000**	**0.000**	0.056	1.000
GPT-5	Anti-Consensus+Reasoned	18	0.400	**0.000**	**0.000**	0.000	1.000

Table 2: Paired robustness tests (`UNS_hit`). No comparison remains significant after Holm correction.

Model	Comparison	Δ`UNS_hit`	Permutation p	Holm-adjusted p	Cohen's d
GPT-4.1	Baseline vs Anti-Consensus	-0.020	0.826	0.826	-0.052
GPT-4.1	Baseline vs Anti-Consensus+Reasoned	+0.107	0.141	0.492	+0.358
GPT-5	Baseline vs Anti-Consensus	-0.047	0.347	0.694	-0.228
GPT-5	Baseline vs Anti-Consensus+Reasoned	-0.068	0.123	0.492	-0.387

4 Results

Main quantitative findings. Table 1 shows clear directional changes for GPT-4.1. Moving from baseline to anti-consensus+reasoned reduces overlap-hit rate from 0.389 to 0.111 and increases mean `UNS_hit` from 0.562 to 0.669 (+0.107). Baseline to anti-consensus also reduces overlap (0.389 to 0.222), but with a slight mean `UNS_hit` decrease (-0.020).

Statistical tests. Table 2 reports paired permutation tests with Holm correction. The largest positive effect is GPT-4.1 baseline vs anti-consensus+reasoned (Cohen's $d = +0.358$), but adjusted $p = 0.492$, so it is not statistically significant at $\alpha = 0.05$. No comparison survives correction.

Model-specific robustness differences. GPT-5 baseline produces usable outputs (match rate 0.667, parse error 0.167), while both anti-consensus variants collapse in structured extraction (parse error 1.000). This failure mode dominates downstream metrics and explains why GPT-5 overlap-hit rates appear low yet `UNS_hit` does not improve.

Ablation-style interpretation across prompt conditions. Treating prompt condition as an intervention ablation, anti-consensus instructions reduce social obviousness for GPT-4.1 but do not guarantee higher composite novelty. Adding reasoned justification improves `UNS_hit` directionally and yields the best overlap reduction for GPT-4.1, suggesting that anti-consensus guidance plus rationale constraints is the strongest of the tested prompt variants.

5 Discussion

What we learn from directional but non-significant gains. The experiment supports the hypothesis in direction but not in corrected significance. For GPT-4.1, anti-consensus prompting lowers overlap with crowd discourse, which is the core behavior we seek to measure. However, small sample size (18 outputs per condition), noisy consensus proxies, and metric variance limit inferential strength.

Proxy quality is the central bottleneck. Mined subreddit overlap captures social obviousness only imperfectly. Title ambiguity, retrieval limits, and false positives from naive mining can inflate or obscure overlap. Our direct title-hit robustness signal is more interpretable, but still not a gold-standard annotation of "obvious vs genuinely underrated."

Evaluation validity requires output reliability. The GPT-5 anti-consensus runs show that prompt-format incompatibility can dominate novelty conclusions. If structured extraction fails, novelty metrics become artifacts of parsing and matching. This reinforces that benchmark validity depends on robust output controls (schema/tool-calling, retries, and strict validation).

Broader implications. A well-calibrated underratedness diagnostic could become a useful stress test for ideation-oriented LLM use, similar in spirit to serendipity tests in recommender systems [Xi

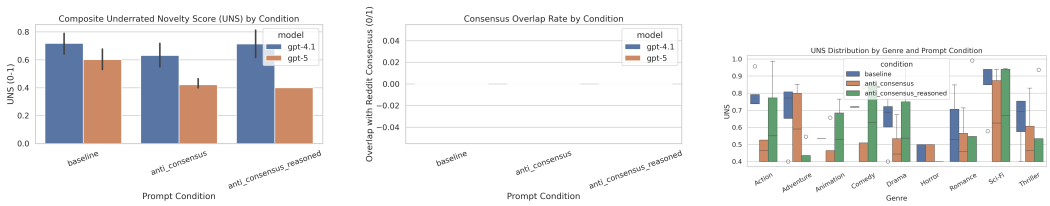

Figure 1: Robustness visualizations. Left: `UNS_hit` by model and condition. Middle: overlap behavior by condition. Right: genre-level variability of `UNS_hit`. Together, the plots show directional overlap reduction for GPT-4.1 under anti-consensus prompting, alongside substantial variance and non-significant corrected tests.

et al., 2025]. Our results suggest that anti-consensus prompting can help, but benchmark design must separate social popularity, quality, and extraction reliability.

Limitations. We use MOVIELENS-LATEST-SMALL small, which misses many recent and long-tail titles. We run only two repetitions per cell, limiting power. We focus on movies only, so generalization to books, music, tools, or scientific ideas remains open. Finally, we do not include human annotation in this pilot, so the overlap proxy is not fully validated.

6 Conclusion

We presented UNDERRATED NOVELTY DIAGNOSTIC, a reproducible evaluation for "most underrated X" prompting that combines data-grounded underratedness with social-consensus overlap and robustness checks. Across two models, nine genres, and three prompt conditions, we find that anti-consensus prompting can reduce obviousness, most clearly for GPT-4.1 (overlap-hit 38.9% baseline to 11.1% in anti-consensus+reasoned). However, corrected statistical tests do not confirm significant `UNS_hit` gains in this pilot setup.

The key takeaway is that "most underrated" prompts are promising diagnostics, but measurement quality determines conclusion quality. Future work should add human-labeled obviousness judgments at scale, expand coverage with larger catalogs (e.g., MovieLens-25M plus metadata linkage), and enforce schema-level output constraints to prevent parsing artifacts. We also plan to test cross-domain transfer beyond movies to evaluate whether the same anti-consensus patterns hold in other discovery tasks.

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